(b) (i) Answer the following giving reason :

(II) A low voltage supply from which one needs high currents must have very low internal resistance. Why ?

CBSE 26

reach its steady value.

(II) $I = \dfrac{\varepsilon}{R + r}$

The maximum current that can be drawn from a cell is for R=0 and it is $I_{max} = \dfrac{\varepsilon}{r}$ 1

(b) A battery of emf 21 V and internal resistance 3 Ω is connected to a resistor. If the current in the circuit is 3 A, find : *2*

(i) the resistance of the resistor.

(ii) the terminal voltage of the battery.

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(b) A battery of emf 21 V and internal resistance 3 Ω is connected to a resistor. If the current in the circuit is 3 A, find : *2*

(i) the resistance of the resistor.

(ii) the terminal voltage of the battery.

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i)	$IR = \varepsilon - Ir$	½
	$R = \dfrac{21-(3\times 3)}{3}$	
	$= 4\Omega$	½
ii)	$V = IR$	½
	$= 3 \times 4$	
	$= 12\ V$	½

17. A battery of emf 6 V and internal resistance 1 Ω is connected to a resistor. The current in the circuit is 0.5 A. Calculate **2**

(i) the resistance of the resistor and

(ii) the terminal voltage of the battery.

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17. A battery of emf 6 V and internal resistance 1 Ω is connected to a resistor. The current in the circuit is 0.5 A. Calculate **2**

(i) the resistance of the resistor and

(ii) the terminal voltage of the battery.

CBSE 26

(a) $E = 6V$, $r = 1\Omega$, $I = 0.5A$	
$V = E - Ir$	½
$IR = E - Ir$	
$R = \frac{E-Ir}{I} = \frac{6-(0.5)(1)}{(0.5)}$	
$R = 11\ \Omega$	½
(b) $V = IR$	½
$V = (0.5)(11)$	
$V = 5.5$ volt	½
Alternatively, $V = E - Ir$	
$= 6 - (0.5)(1)$	
$= 5.5$volt	

19. The terminal potential difference of a cell is 19 V when a current of 1·0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3·0 A. Find the emf and internal resistance of the cell. *2*

Finding the emf	1
Finding the internal resistance	1

19. The terminal potential difference of a cell is 19 V when a current of 1·0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3·0 A. Find the emf and internal resistance of the cell. *2*

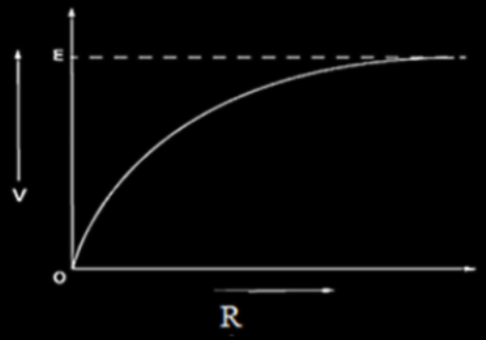
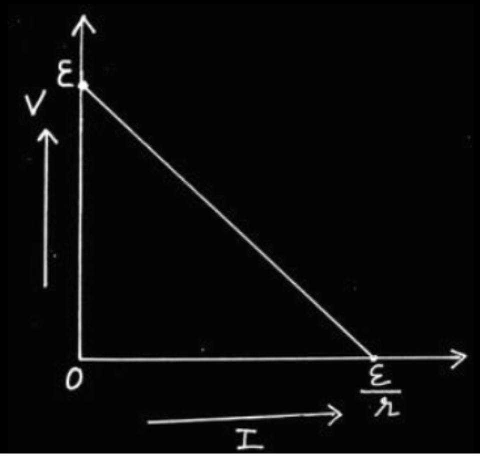
$V = E - Ir$	½
$19 = E - 1.r$ ____________(i) $17 = E - 3.r$ ____________(ii)	½
$2r = 2 \Rightarrow r = 1\,\Omega$	½
$E = 19+1 = 20V$	½

Finding the emf	1
Finding the internal resistance	1

22. A cell of emf E and internal resistance r is connected to a variable resistance R. Draw plots showing the variation of (a) terminal voltage V with R, and (b) V with current I, in the circuit. *2*

Plots of	
(a) Terminal voltage with R	1
(b) V with current I	1

22. A cell of emf E and internal resistance r is connected to a variable resistance R. Draw plots showing the variation of (a) terminal voltage V with R, and (b) V with current I, in the circuit. *2*

(a)		**1**
(b)		**1**

Plots of	
(a) Terminal voltage with R	1
(b) V with current I	1

20. Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell ? *2*

Two differences	½+ ½
Important precaution	1

20. Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell ? *2*

Two differences 1. The potential difference across the electrodes in open circuit is e.m.f. (ε) and in closed circuit is terminal potential difference (V). 2. V depends on r and ε is independent of r.	½ ½
Precaution- 1. Some external resistance should be connected to cell in series. 2. Short circuiting should be avoided.	1

Two differences	½+ ½
Important precaution	1

1. A battery of emf E and internal resistance r is connected to an external circuit. The potential drop within the battery is proportional to :

(a) current in the circuit

(b) total resistance of the circuit

(c) emf of the battery

(d) power dissipated in the circuit

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(c) emf of the battery

(d) power dissipated in the circuit

(a) Current in the circuit.

17. *Assertion (A) :* The internal resistance of a cell is constant.

Reason (R) : Ionic concentration of the electrolyte remains same during use of a cell.

17. *Assertion (A) :* The internal resistance of a cell is constant.

Reason (R) : Ionic concentration of the electrolyte remains same during use of a cell.

(d) Assertion (A) is false and Reason (R) is also false

13. The emf and internal resistance of a cell are E and r respectively. It is connected across an external resistance R = 2r. The potential drop across the terminals of the cell will be :

(a) $\frac{E}{4}$

(b) $\frac{E}{2}$

(c) $\frac{2}{3}E$

(d) $\frac{E}{3}$

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13. A cell of emf E is connected across an external resistance R. When current 'I' is drawn from the cell, the potential difference across the electrodes of the cell drops to V. The internal resistance 'r' of the cell is **1**

(a) $\left(\frac{E-V}{E}\right)R$

(b) $\left(\frac{E-V}{R}\right)$

(c) $\frac{(E-V)\,R}{I}$

(d) $\left(\frac{E-V}{V}\right)R$

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(c) $\frac{(E-V)R}{I}$

(d) $\left(\frac{E-V}{V}\right)R$

(d) $\left(\frac{E-V}{V}\right)R$

OR

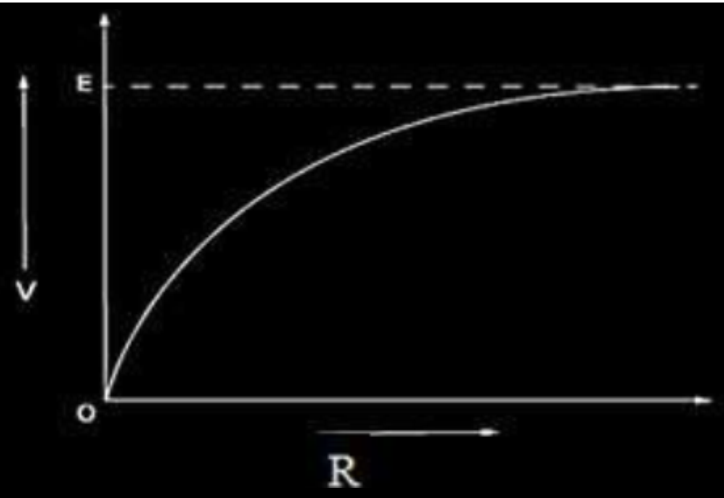
(b) (i) A cell emf of (E) and internal resistance (r) is connected across a variable load resistance (R). Draw plots showing the variation of terminal voltage V with (i) R and (ii) the current (I) in the load. **2**

(ii) Three cells, each of emf E but internal resistances 2r, 3r and 6r are connected in parallel across a resistor R.

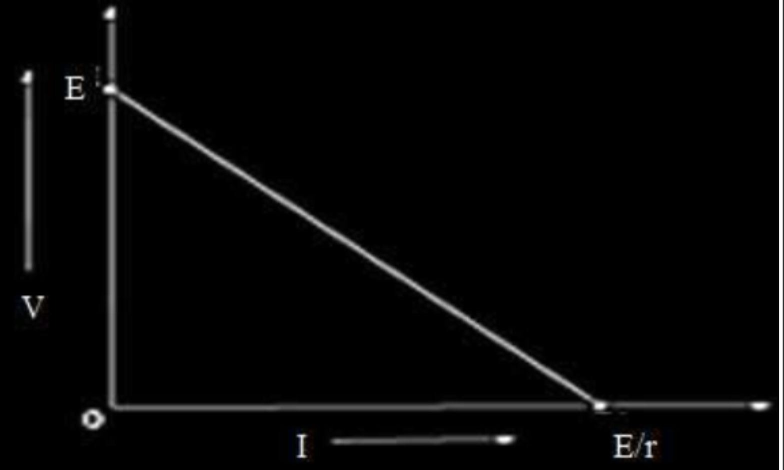
Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell. **3**

(i)
i Plot showing variation of V with R (1)
ii Plot showing variation of V with I (1)
(ii)
(i) Obtaining expression for current flowing through circuit $(1\frac{1}{2})$
(ii) Obtaining expression for terminal potential difference across the equivalent cell. $(1\frac{1}{2})$

(i) i.



ii.



(ii)

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

$$\frac{1}{r_{eq}} = \frac{1}{2r} + \frac{1}{3r} + \frac{1}{6r}$$

$$r_{eq} = r \qquad \text{½}$$

Given cells are of equal emf (E) and connected in parallel, so

$$\frac{E_{eq}}{r_{eq}} = \frac{E}{2r} + \frac{E}{3r} + \frac{E}{6r}$$

$$E_{eq} = E \qquad \text{½}$$

Current flowing $I = \dfrac{E_{eq}}{R + r_{eq}}$

$$I = \frac{E}{R + r} \qquad \text{½}$$

The terminal potential difference

$$V = E_{eq} - Ir_{eq} \qquad \text{½}$$

$$V = E - \frac{E}{(R + r)} \times r \qquad \text{½}$$

$$V = \frac{ER}{R + r} \qquad \text{½}$$

1. A battery supplies 0.9 A current through a 2 Ω resistor and 0.3 A current through a 7 Ω resistor when connected one by one. The internal resistance of the battery is :

(A) 2 Ω

(B) 1.2 Ω

(C) 1 Ω

(D) 0.5 Ω

1. A battery supplies 0.9 A current through a 2 Ω resistor and 0.3 A current through a 7 Ω resistor when connected one by one. The internal resistance of the battery is :

(A) 2 Ω
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(D) 0.5 Ω

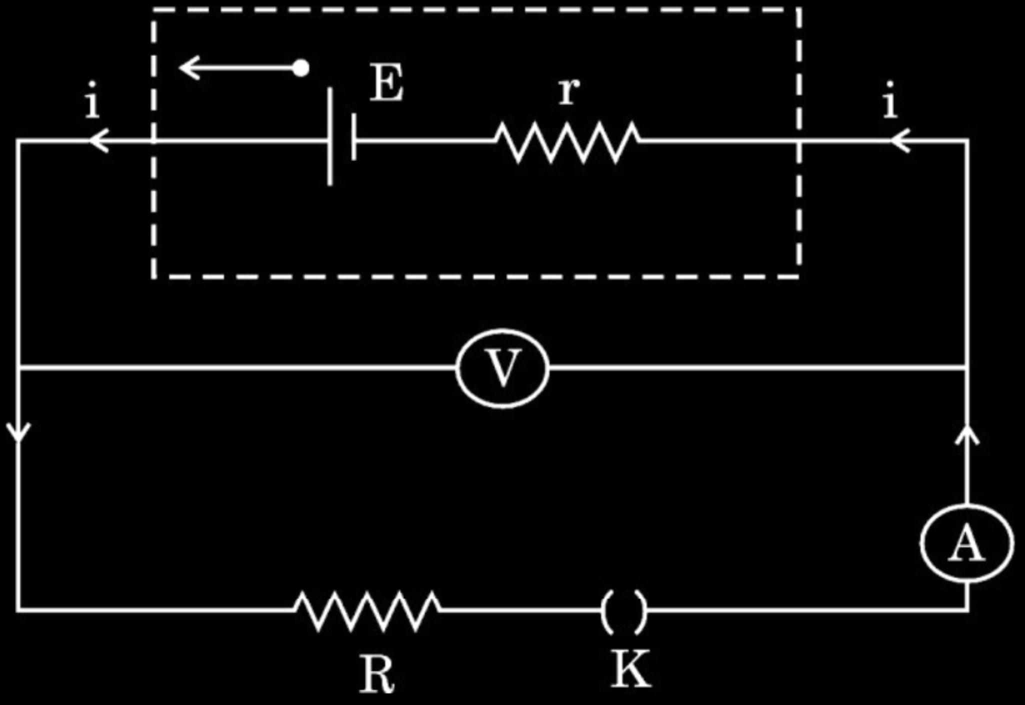
(D) 0.5 Ω

22. A battery of unknown emf E and internal resistance r is connected in a circuit as shown in the figure. When the key (K) is open, the voltmeter reads 10·0 V and ammeter reads 0 A. In the closed circuit, the voltmeter reads 6·0 V and ammeter reads 2·0 A. Calculate : *3*

(a) emf of the battery,

(b) internal resistance of the battery (r), and

(c) external resistance (R).



Calculation of	
(a) emf of battery	½
(b) Internal resistance of battery(r)	1½
(c) external resistance (R)	1

(a) $V = \mathit{E} = 10$ V(When key K is open and $I=0$ A)	½
(b) $V=\mathit{E}-Ir$ (When key K is closed and $I=2$ A)	½
$6=10-2r$	½
$r = 2\Omega$	½
(c) $E=I(r+R)$	½
$10=2(2+R)$	
$R= 3\ \Omega$	½

30. When the terminals of a cell are connected to a conductor of resistance R, an electric current flows through the circuit. The electrolyte of the cell also offers some resistance in the path of the current, like the conductor. This resistance offered by the electrolyte is called internal resistance of the cell (r). It depends upon the nature of the electrolyte, the area of the electrodes immersed in the electrolyte and the temperature. Due to internal resistance, a part of the energy supplied by the cell is wasted in the form of heat.

When no current is drawn from the cell, the potential difference between the two electrodes in known as emf of the cell (ε). With a current drawn from the cell, the potential difference between the two electrodes is termed as terminal potential difference (V).

(i) Choose the ***incorrect*** statement : *1*

(A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf (ε), during discharge of the cell.

(B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.

(C) When current is drawn from the cell then $V = \varepsilon - Ir$.

(D) The graph between potential difference between the two terminals of the cell (V) and the current (I) through it is a straight line with a negative slope.

(ii) Two cells of emfs 2·0 V and 6·0 V and internal resistances 0·1 Ω and 0·4 Ω respectively, are connected in parallel. The equivalent emf of the combination will be : *1*

(A) 2·0 V (B) 2·8 V

(C) 6·0 V (D) 8·0 V

(i) Choose the ***incorrect*** statement : *1*

(A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf (ε), during discharge of the cell.

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(A) 2·0 V

(B) 2·8 V

(C) 6·0 V

(D) 8·0 V

(iii) Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential V_+ ($V_+ > 0$), and the negative electrode develops a potential $-(V_-)$ ($V_- \geq 0$), relative to the electrolyte adjacent to it. When no current is drawn from the cell then : *1*

(A) $\varepsilon = V_+ + V_- > 0$ (B) $\varepsilon = V_+ - V_- > 0$

(C) $\varepsilon = V_+ + V_- < 0$ (D) $\varepsilon = V_+ + V_- = 0$

(iv) (a) Five identical cells, each of emf 2 V and internal resistance 0·1 Ω are connected in parallel. This combination in turn is connected to an external resistor of 9·98 Ω. The current flowing through the resistor is : *1*

(A) 0·05 A (B) 0·1 A

(C) 0·15 A (D) 0·2 A

OR

(b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is : *1*

(A) 1·0 Ω (B) 1·5 Ω

(C) 2·0 Ω (D) 2·5 Ω

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(C) 0·15 A $\boxed{\text{(D)} \quad 0{\cdot}2 \text{ A}}$

OR

(b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is : *1*

$\boxed{\text{(A)} \quad 1{\cdot}0\ \Omega}$ (B) 1·5 Ω

(C) 2·0 Ω (D) 2·5 Ω

2. The emf and internal resistance of a cell are 3 V and $0{\cdot}2\ \Omega$ respectively. It is connected to an external resistor of $5{\cdot}8\ \Omega$. The potential difference across the cell will be :

(A) 1·2 V

(B) 2·0 V

(C) 2·9 V

(D) 3·2 V

2. The emf and internal resistance of a cell are 3 V and $0{\cdot}2\ \Omega$ respectively. It is connected to an external resistor of $5{\cdot}8\ \Omega$. The potential difference across the cell will be :

(A) 1·2 V (B) 2·0 V

(C) 2·9 V (D) 3·2 V

(C) 2.9 V

19. (a) A cell is connected across an external resistance 12 Ω and supplies 0.25 A current. When the external resistance is increased by 4 Ω, the current reduces to 0.2 A. Calculate (i) the emf, and (ii) the internal resistance, of the cell. *2*

19. (a) A cell is connected across an external resistance 12 Ω and supplies 0.25 A current. When the external resistance is increased by 4 Ω, the current reduces to 0.2 A. Calculate (i) the emf, and (ii) the internal resistance, of the cell. *2*

(i) Calculating e.m.f of cell	1
(ii) Calculating internal resistance of cell	1

$$I = \frac{\varepsilon}{R+r}$$

$$0.25 = \frac{\varepsilon}{12+r} \qquad ------(1)$$ ½

$$0.2 = \frac{\varepsilon}{16+r} \qquad --------(2)$$ ½

On solving eq (1) and (2)

$r = 4\Omega$ ½

$\varepsilon = 4\,\text{V}$ ½

17. A cell of emf E and internal resistance r is connected across a resistor of variable resistance R. Show graphically the variation of

(a) the terminal voltage across the cell,

(b) the current supplied by the cell,

with R as it is increased from 0 to the maximum value. *2*

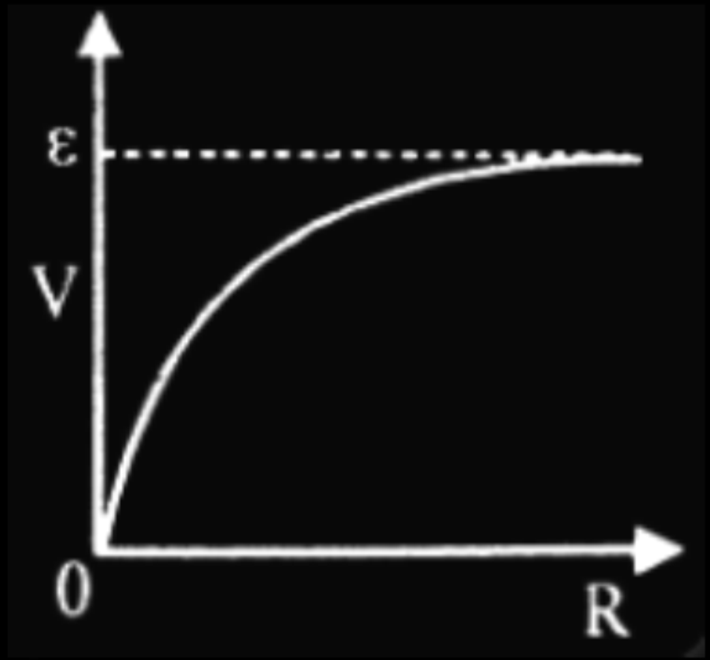
17. A cell of emf E and internal resistance r is connected across a resistor of variable resistance R. Show graphically the variation of

(a) the terminal voltage across the cell,

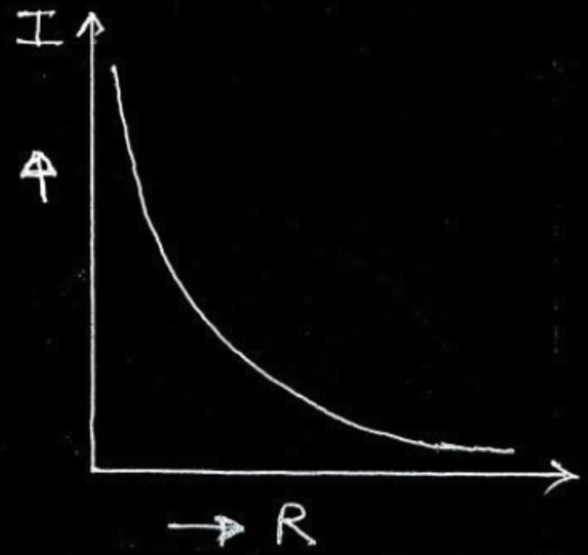
(b) the current supplied by the cell,

with R as it is increased from 0 to the maximum value. *2*

(a)

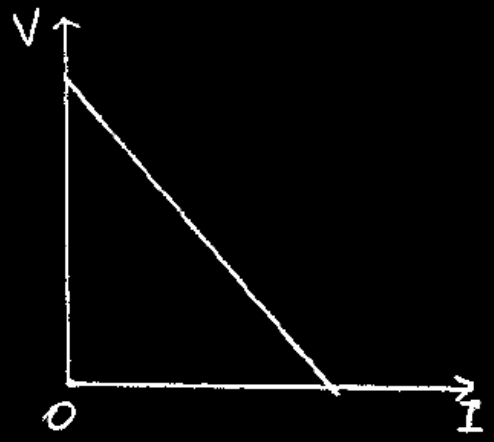


(b)



17. A cell of emf E and internal resistance r is connected to an external variable resistance R. Plot a graph showing the variation of terminal voltage V of the cell as a function of current I, supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it. *2*

17. A cell of emf E and internal resistance r is connected to an external variable resistance R. Plot a graph showing the variation of terminal voltage V of the cell as a function of current I, supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it. *2*

	1
$V = E - Ir$	
E = intercept on y-axis (i.e. V-axis)	½
r = |slope of graph|	½

22. (a) Define the terms 'emf' and 'terminal voltage' of a cell. Can the terminal voltage of a cell ever exceed its emf ? Explain. **3**

OR

(b) Define the term current density. Write its SI unit. Derive the equivalent form of Ohm's law $\vec{j} = \sigma\vec{E}$. **3**

• Defining emf and terminal voltage	1+1
• Explaining terminal voltage exceeding the emf	1

Defining current density and Writing its SI unit	1+ ½
Deriving $\vec{J} = \sigma\vec{E}$	1 ½

emf- It is the potential difference across the terminal of the cell, when no current is being drawn from it. **Terminal Voltage** - It is the potential difference between the terminals of the cell in a closed circuit. **Alternatively** $V = \varepsilon - Ir$ Yes, During charging, $V = \varepsilon - (-I)\,r = \varepsilon + Ir$	1 1 ½ ½
Current density – It is defined as the current through unit area normal to the current. **Alternatively**, $\vec{J} = \frac{I}{\vec{A}}$ S.I unit A/m^2 **Derivation** $V = IR$ $\vec{E}l = I\frac{\rho l}{\vec{A}}$ $\vec{E} = \vec{J}\rho$ $\vec{J} = \frac{1}{\rho}\vec{E} = \sigma\vec{E}$	1 ½ ½ ½ ½

Alternatively

$$I = ne\vec{A}v_d$$

$$\frac{I}{\vec{A}} = \frac{ne^2\tau\vec{E}}{m}$$

$$\vec{J} = \frac{\vec{E}}{\rho} = \sigma\vec{E}$$

OR

(b) (i) A battery of emf E and internal resistance r is connected to a variable external resistance R.

(I) Obtain the expression for current I in the circuit and the value of maximum current the battery can supply.

(II) Obtain the terminal voltage V across the battery and its maximum possible value.

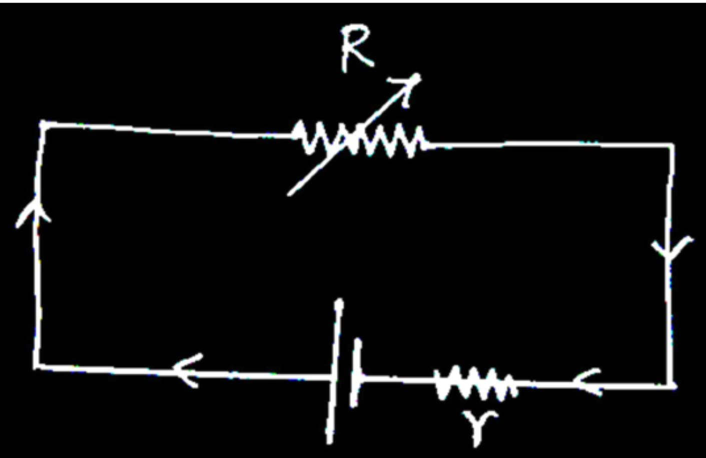
(ii) The above battery sends a current I_1 when $R = R_1$ and a current I_2 when $R = R_2$. Obtain the internal resistance of the battery in terms of I_1, I_2, R_1 and R_2. *3*

(i) (I)

- Obtaining the expression for current ½
- Finding value of maximum current ½

(II) Obtaining terminal voltage V and its maximum possible value 1

(ii) Obtaining the internal resistance of the battery 1



- $V = E - Ir$

  $IR = E - Ir$

  $I = \dfrac{E}{R+r}$ ½

  For maximum value of current R=0

$I_{max} = \dfrac{E}{r}$ ½

(II) $V = V_+ + V_- - Ir$

$V = E - Ir$ ½

$V_{max} = E$, when I=0 ½

(ii) $I_1R_1 + I_1r = I_2R_2 + I_2r$ ½

$r = \dfrac{I_2R_2 - I_1R_1}{I_1 - I_2}$ ½

2. A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is **1**

(A) 1.05 (B) 1

(C) 0.95 (D) 1.1

2. A battery of e.m.f. 12 V and internal resistance 0.5 Ω is connected to a 9.5 Ω resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is 1

$\boxed{\text{(A)} \quad 1.05}$ (B) 1

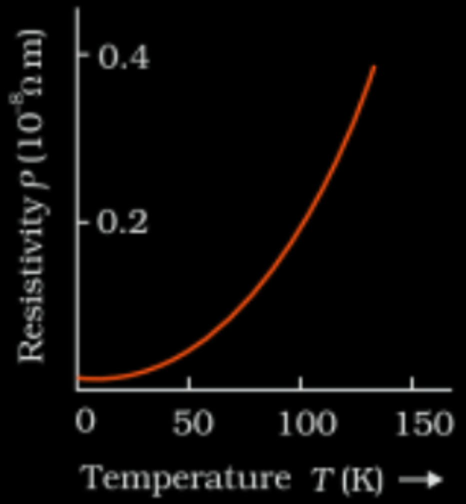
(C) 0.95 (D) 1.1

22. (a) Define resistivity of a conductor. Discuss its dependence on temperature of the conductor and draw a plot of resistivity of copper as a function of temperature. **3**

(b) (i) "A low voltage battery from which high current is required must have low internal resistance." Justify.

(ii) "A high voltage battery must have a large internal resistance." Justify.

(a) Defining resistivity	1
Discussing its dependence on temperature	½
Plotting graph of resistivity with temperature for copper	½
(b) (i) Justification	½
(ii) Justification	½

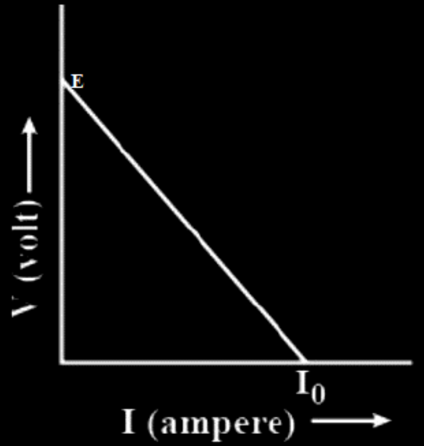
(a) Resistivity is the resistance of a material of unit length having unit area of cross-section. On increasing the temperature of a conductor, the resistivity increases.  **Note**: Full credit to be given if values are not shown on the graph.	1 ½ ½
(b) (i) A low internal resistance allows large current to be drawn even at a low voltage. (ii) To limit the current.	½ ½

22. (a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop V across R. Obtain the relationship between V, E, R and r. **3**

(b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?

22. (a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop *V* across R. Obtain the relationship between *V*, E, R and r. **3**

(b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?

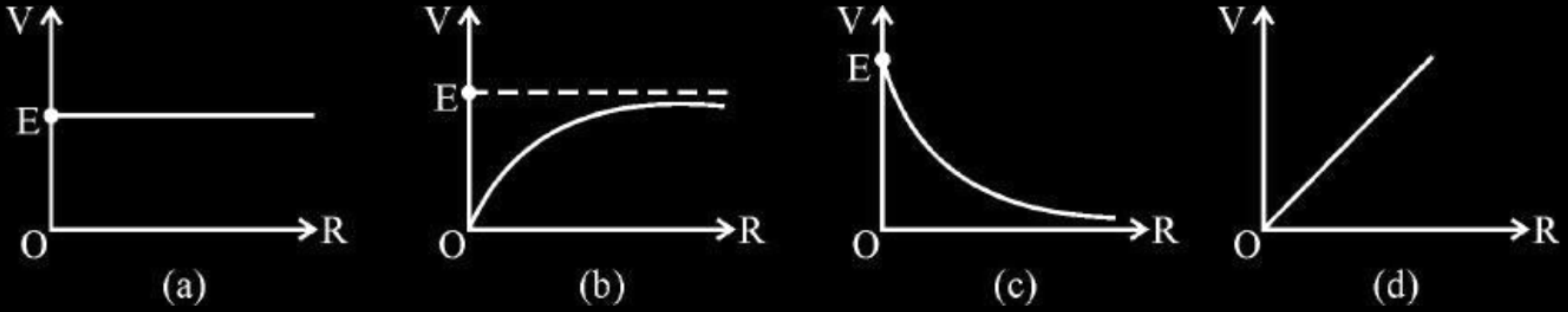
(a) Current drawn from the cell $(I) = \dfrac{E}{r+R}$	½
Potential difference (V)=IR	
$V = \dfrac{ER}{r+R}$	½
(b) 	1
Slope of the graph is equal to internal resistance of the cell.	½
The intercept on Y-axis gives the emf of the cell.	½

17. A battery of emf E and internal resistance r is connected to a rheostat. When a current of 2A is dawn from the battery, the potential difference across the rheostat is $5V$. The potential difference becomes $4V$ when a current of 4A is drawn from the battery. Calculate the value of E and r. **2**

17. A battery of emf E and internal resistance r is connected to a rheostat. When a current of 2A is dawn from the battery, the potential difference across the rheostat is $5V$. The potential difference becomes $4V$ when a current of 4A is drawn from the battery. Calculate the value of E and r. **2**

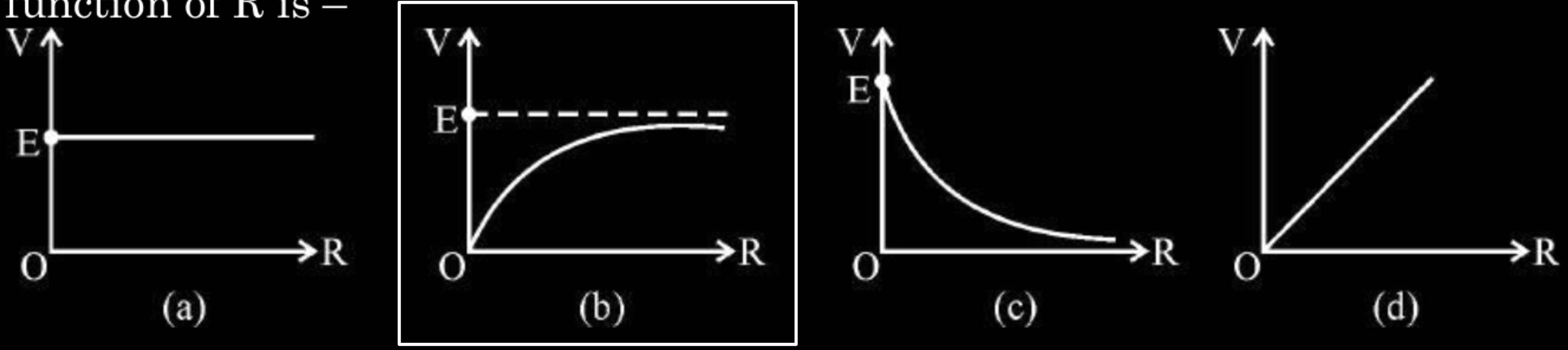
Calculating the value of E 1 Calculating the value of r 1 $E = V + Ir$ In first case $E = 5 + 2r$ In second case $E = 4 + 4r$ After solving	½ ½
E= 6 V r = 0.5 Ω	½ ½

1. A cell of emf (E) and internal resistance r is connected across a variable external resistance R. The graph of terminal potential difference V as a function of R is –



1

1. A cell of emf (E) and internal resistance r is connected across a variable external resistance R. The graph of terminal potential difference V as a function of R is –



(a) (b) (c) (d) 1

1. A cell of internal resistance r connected across an external resistance R can supply maximum current when *1*

(A) $R = r$

(B) $R > r$

(C) $R = \frac{r}{2}$

(D) $R = 0$

1. A cell of internal resistance r connected across an external resistance R can supply maximum current when *1*

(A) $R = r$

(B) $R > r$

(C) $R = \frac{r}{2}$

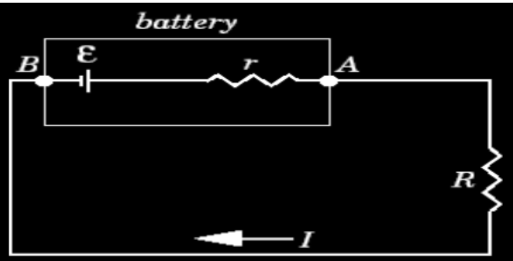
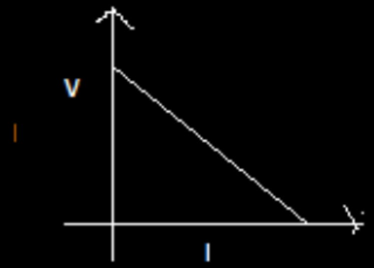
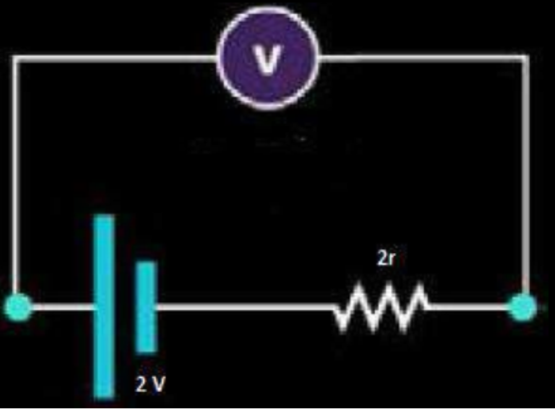
$\boxed{\text{(D) } R = 0}$

CBSE 2019

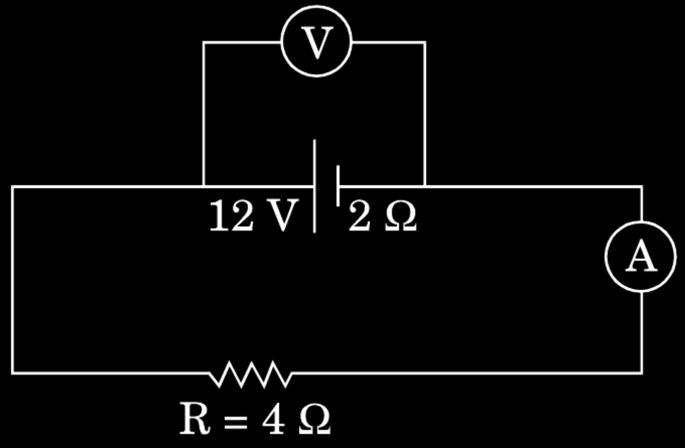
(a) Derive a relation between the internal resistance, emf and terminal potential difference of a cell from which current I is drawn. Draw V vs I graph for a cell and explain its significance.

(b) A voltmeter of resistance 998 Ω is connected across a cell of emf 2 V and internal resistance 2 Ω. Find the potential difference across the voltmeter and also across the terminals of the cell. Estimate the percentage error in the reading of the voltmeter.

a) Relation between E, V, r	1
Graph V v/s I	1
Significance of graph	1
b) Current in voltmeter	1
Potential difference across voltmeter	½
Percentage Error	½

(a)	 $E- IR - rI = 0$ $E - V - Ir = 0$ $E = V + Ir$ [Award 1 mark even if student writes the relation directly]  Significance of Graph – To find cmf and internal resistance of cell.	½ ½ 1
(b)	$V = E - Ir$ $998 \times I = 2 - 2I$ $1000 \times I = 2$ $I = \frac{2}{1000} = .002\ A$ $V = .002 \times 998$ $V = 1.996\ V$ $\%\ error = \frac{.004}{2} \times 100 = 0.2\ \%$  	½ ½ ½

(b) In the figure shown, an ammeter A and a resistor of 4 Ω are connected to the terminals of the source. The emf of the source is 12 V having an internal resistance of 2 Ω. Calculate the voltmeter and ammeter readings.



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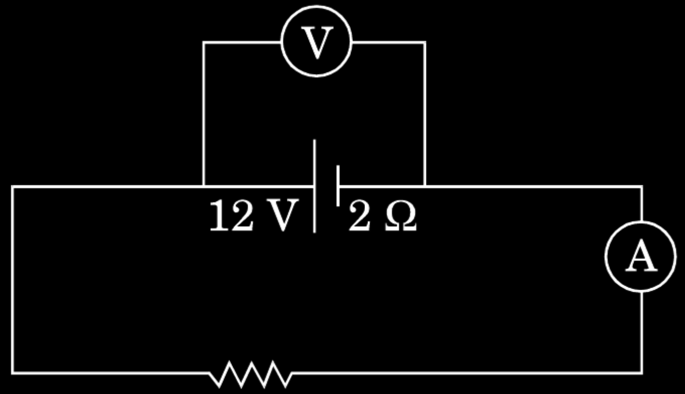
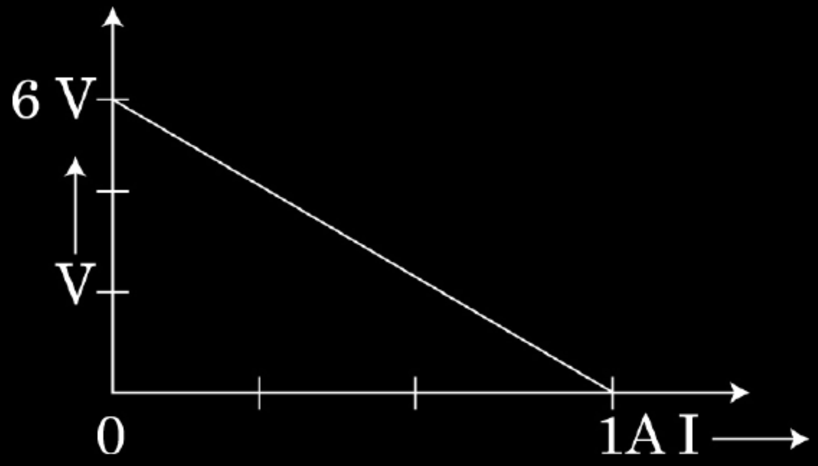
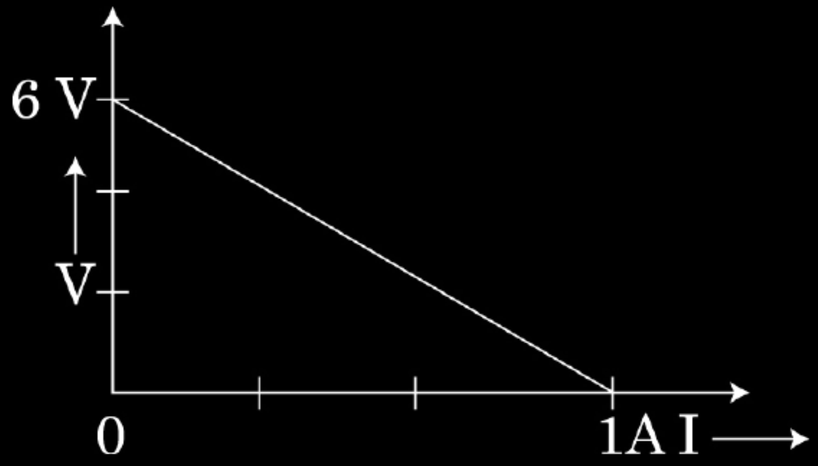


b) Ammeter Reading $I = \frac{V}{R+r}$	½
$= \frac{12}{4+2} A = 2A$	½
Voltmeter Reading $V = E - Ir$	½
$= [12 - (2 \times 2)]\ V = 8V$ (Alternatively, $V = iR = 2 \times 4V = 8V$)	½

The plot of the variation of potential difference across a combination of three identical cells in series, versus current is shown below. What is the emf and internal resistance of each cell ?



The plot of the variation of potential difference across a combination of three identical cells in series, versus current is shown below. What is the emf and internal resistance of each cell ?

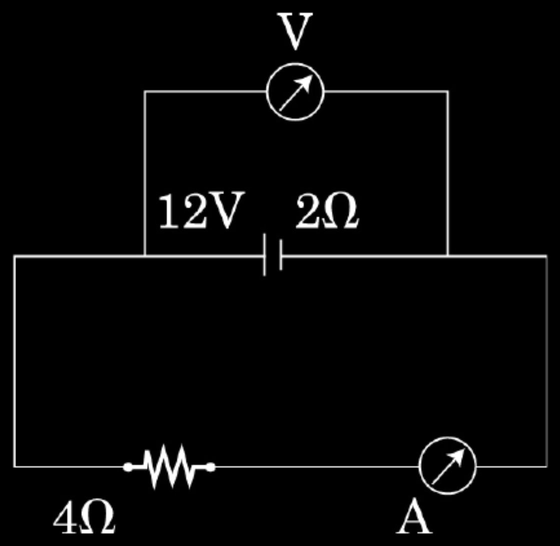


E = 2V	½
$r = 2\Omega$	½

A battery of emf 12V and internal resistance 2 Ω is connected to a 4 Ω resistor as shown in the figure.

(a) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.

(b) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit ?



a) Effective resistance of the circuit R_E= 6Ω $\therefore I = \frac{12A}{6} = 2A$	
Terminal potential difference across the cell, V=E-ir	½
Also p.d. across 4Ω resistor =4X2V= 8V Hence the volmeter gives the same reading in the two cases.	½
b) In series -current same	½
In parallel – potential same	½